# Do Now · retrieval

## Search engine indexing and PageRank

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is a hyperlink?

### Q2

What is an index in retrieval?

### Q3

What kind of graph models links?

### Q4

BST root20, left10, right30; insert 25. Where is it placed?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — A crawler discovers pages by following links and sitemaps; an index records useful information about documents for searching.

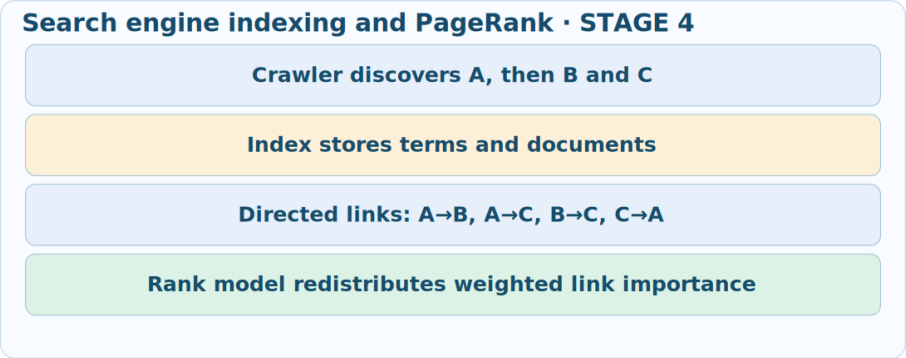
## Core idea 2

Apply — PageRank models a random web surfer and passes importance through directed links. A link from a high-ranked page can carry more weight than one from a low-ranked page.

## Core idea 3

Evaluate — PageRank uses damping and repeated iterations; raw incoming link counts alone are not the full algorithm. Search engines combine many ranking signals.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

crawler — Discovers/fetches documents.

inverted index — Maps terms to documents.

PageRank — Iterative link-based importance model.

damping — Probability factor in random-surfer model.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

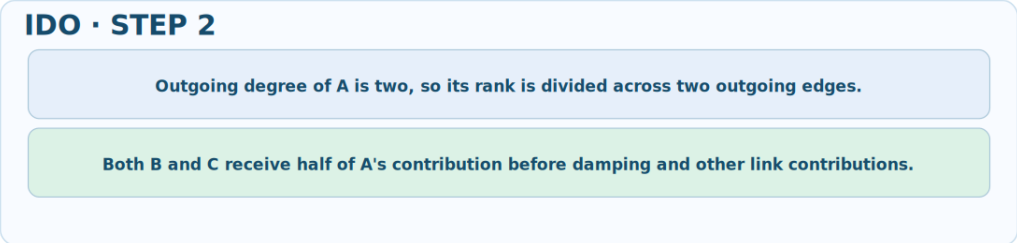
In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

## Follow the worked method

1. Predict first — Outgoing degree of A is two, so its rank is divided across two outgoing edges.

2. Explain — Both B and C receive half of A's contribution before damping and other link contributions.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first

The outgoing links split A's contribution evenly in the simplified model.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Distinguish crawling and indexing. [4]

## Question 2

In directed link graph A→B,C and B→C, identify incoming links of C and A's share given rank 0.4, ignoring damping. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why the number of inbound links alone is not an adequate PageRank calculation. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

A page's PageRank is simply its number of incoming links.

## Use key vocabulary from memory

1. Explain crawler in your own words.

2. Explain inverted index in your own words.

3. Explain PageRank in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.